

True / False

1. Claim:

$$[\text{matrix unit}]^T = \text{another} [\text{matrix unit}]$$

What is a matrix unit?

$$E_{ik} = [\vec{e}_i]_{m \times 1} \cdot [\vec{e}_k^T]_{1 \times n}$$

↑
outer product

Let $n = 3$. We see

$$E_{11} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow E_{11}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Looks like claim could be true.

Let's check another matrix unit.

$$E_{23} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow E_{23}^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Claim is TRUE.

2. $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $\vec{b} = [b_1 \ b_2 \ \dots \ b_m]$

$$\vec{x} \cdot \vec{b}^T$$

$$= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} [b_1 \ b_2 \ \dots \ b_m]$$

FALSE, but why?

- Are $\vec{x}$ and $\vec{b}$ both column vectors?
- If not, is $\cdot$ an inner product?

3. Let $n = 4$

$$S_{23}(5) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow S_{23}(5)^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow S_{23}(5) = S_{32}(5)$$

TRUE

4. T

5. T

$$6. \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4 \times 4}$$

$$= \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \vec{e}_3 \cdot \vec{e}_3^T$$

TRUE

Multiple Choice

1. $\mathbb{R}^{m \times n}$

$m = \text{rows}$

$n = \text{columns}$

B) 6

2.
$$\begin{bmatrix} 9 & 5 & 3 \\ 8 & 0 & 2 \\ 7 & -6 & 1 \end{bmatrix}_{3 \times 3} - \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 0 & 0 & 0 \\ -10 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 9 & 5 & -3 \\ -2 & 4 & 2 \\ 7 & -6 & 1 \end{bmatrix}$$

E.

3. $A = -2 \cdot \vec{e}_3 \cdot \vec{e}_1^T + 4 \cdot \vec{e}_2 \cdot \vec{e}_2^T + 3 \cdot \vec{e}_3 \cdot \vec{e}_3^T - \vec{e}_1 \cdot \vec{e}_2^T$

$$= -2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$= -2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} + 4 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 3 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 4 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

$$A(:, 2) \cdot A(1, :) = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{D.}$$

$$4. \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 4 \\ 2 & 2 & 0 \\ 3 & 0 & 2 \end{bmatrix}$$

A,